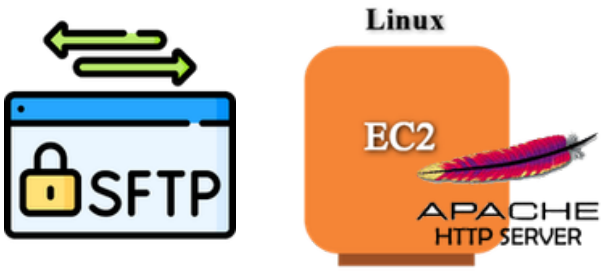
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

7.Configure SFTP toLinux Web Server in AWS



SHAMITHA S
CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.

Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

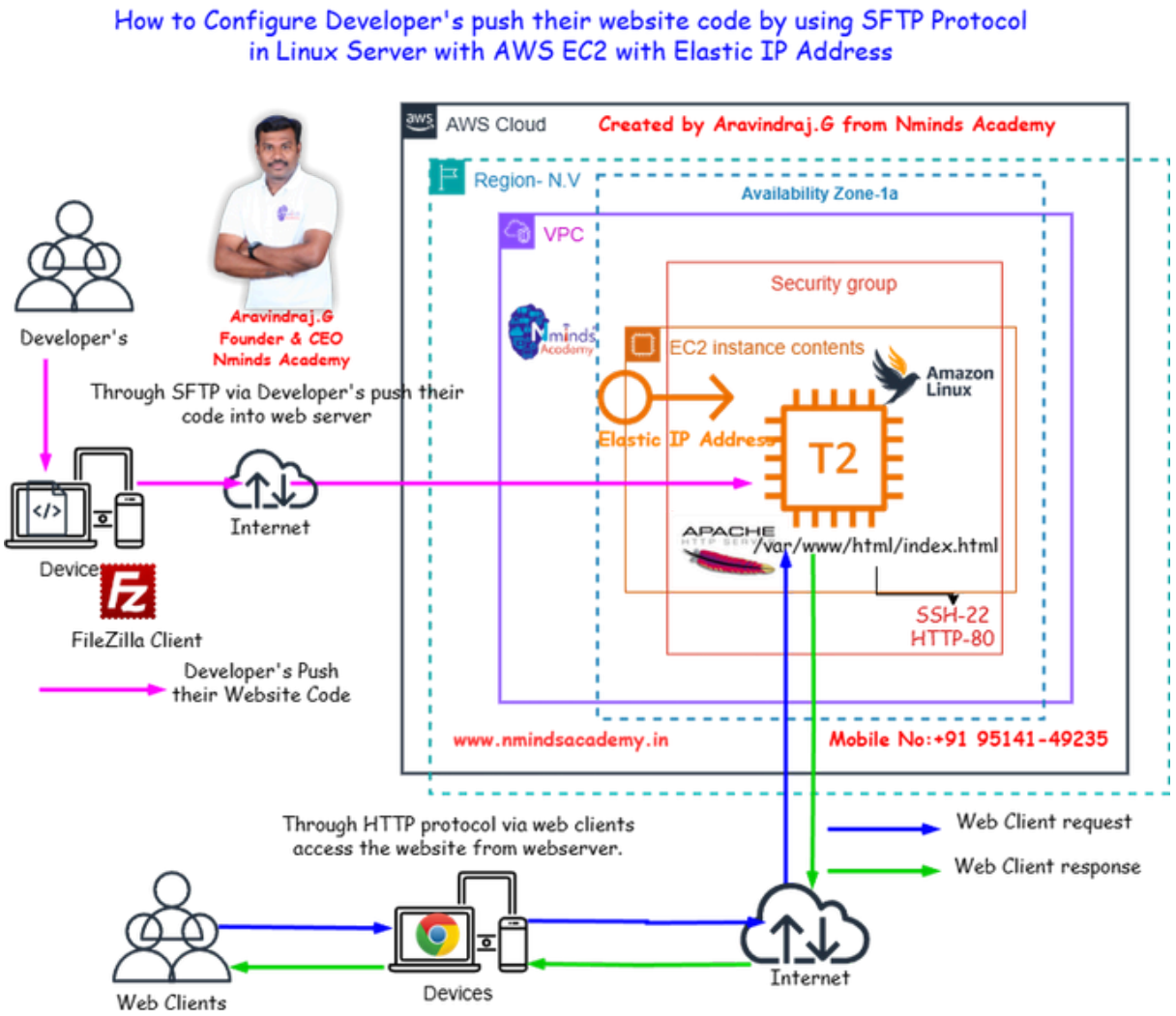
SFTP- Secure File Transfer Protocol

SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

Used Services in Windows Server

Amazon EC2

Amazon Elastic IP Address

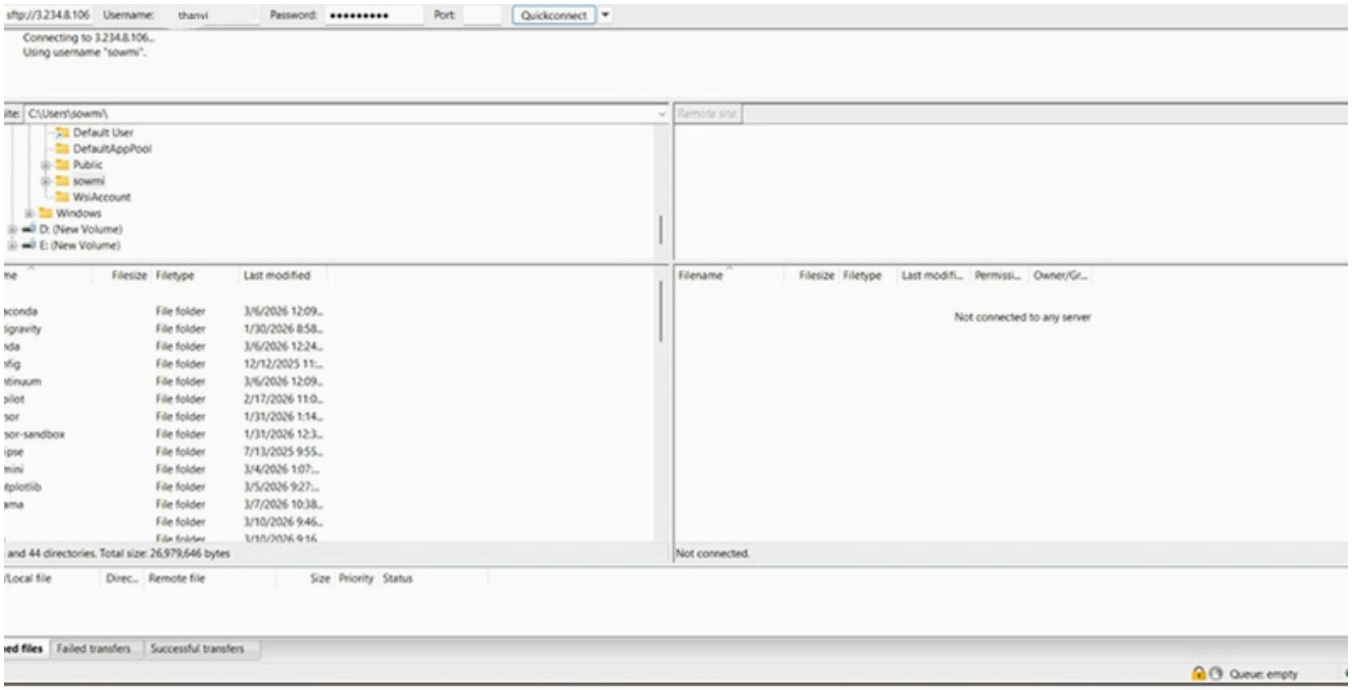
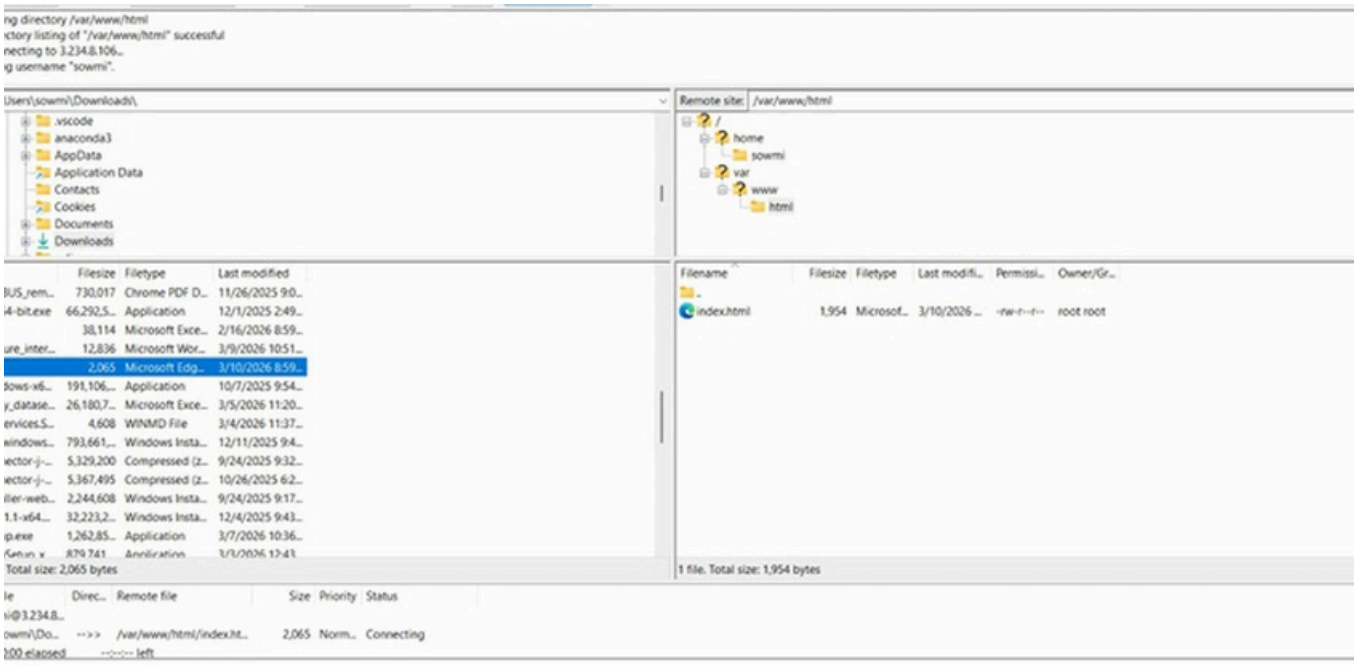
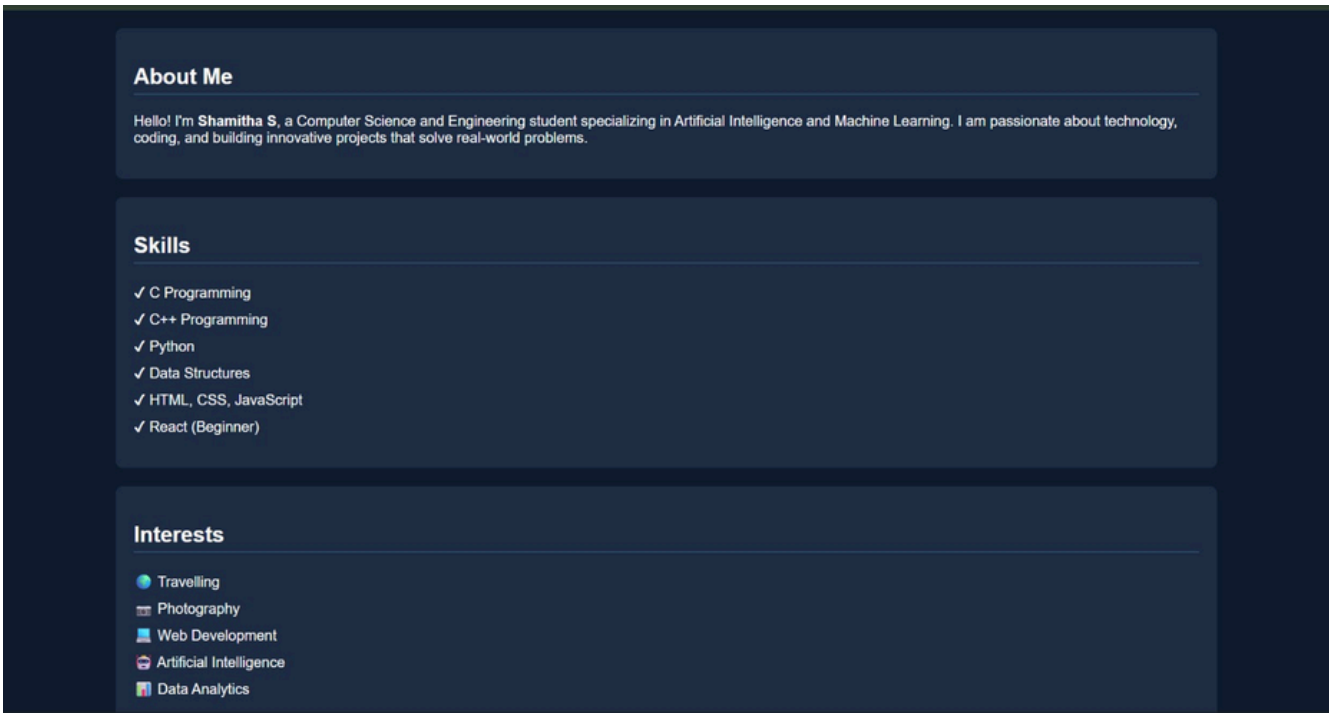
APACHE HTTP SERVER

Apache Webserver

SFTP

Execution Tasks:

```
ng
Facility AUTH
el INFO

ntication:

raceTime 2m
RootLogin prohibit-password
Modes yes
hTries 6
sions 10

uthentication no

efault is to check both .ssh/authorized_keys and .ssh/authorized_keys2
his is overridden so installations will only check .ssh/authorized_keys
zedKeysFile      .ssh/authorized_keys

izedPrincipalsFile none


his to work you will also need host keys in /etc/ssh/ssh_known_hosts
sedAuthentication no
e to yes if you don't trust ~/.ssh/known_hosts for
asedAuthentication
UserKnownHosts no
 read the user's ~/.rhosts and ~/.shosts files
Rhosts yes

citly disable PasswordAuthentication. By presetting it, we
 the cloud-init set_passwords module modifying sshd_config and
rting sshd in the default instance launch configuration.
dAuthentication yes
mptyPasswords no

e to no to disable s/key passwords
eractiveAuthentication yes

ros options
osAuthentication no
RT --                                                          65,27
```



Execution Tasks: